

CHE 422

1. Course number and name

CHE 422. Biochemical Engineering

2. Credits, contact hours and categorization

Credit hours: 3. Contact hours: 3 lecture hours per week.

3. Instructor's or course coordinator's name

Ying Liu

4. Text book, title, author, publisher, year

Bioprocess Engineering: Basic Concepts by Michael L. Shuler, Fikret Kargi, Matthew DeLisa, Prentice Hall, 3rd Edition, 2017.

a. Other supplemental materials

Molecular Biology of the Cell, 6th Edition, Bruce Alberts, Alexander D. Johnson, Julian Lewis, David Morgan, Martin Raff, Keith Roberts, Peter Walter, Garland Science, 2014.

5. Specific course information

a. Brief description of the content of the course (catalog description)

Enzyme-catalyzed and microbially-mediated processes. Free and immobilized enzymes. Batch and continuous cell cultures. Transport phenomena in microbial systems and fermentation processes. Design of biological reactors.

b. Prerequisites or co-requisites

Consent of the instructor.

c. Indicate whether a required, elective, or selected elective.

Elective.

6. Specific goals for the course

a. Specific outcomes of instruction.

To be able to describe and understand recombinant DNA, fermentation and separation methods such as PCR, filtration and centrifugation, cell rupture, and purification techniques such as adsorption, solvent extraction, membrane filtration, chromatography (affinity, ion, hydrophobic, reverse phase), precipitation, and crystallization. To apply fundamental principles for the separation of biopharmaceuticals via chemical engineering unit operations.

b. Explicitly indicate which of the student outcomes are addressed by the course.

Student Learning Outcomes		1	2	3	4
(1)	an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.		x		
(2)	an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.			x	
(3)	an ability to communicate effectively with a range of audiences.			x	
(4)	an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	x			
(5)	an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.	x			
(6)	an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	x			
(7)	an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.		x		

7. Brief list of topics to be covered

The course is aimed at senior undergraduate and graduate students and is designed to move engineers and other non-biologists past the initial language and knowledge barriers of the Biochemical Engineering field.

- Fundamental recombinant DNA techniques
- Genetically engineered biomacromolecules,
- General molecular biology techniques; Polymerase Chain Reaction (PCR) basics, site-directed and random mutagenesis, display techniques, DNA sequencing, DNA and protein gel electrophoresis, cutting and splicing DNA, protein expression, DNA and RNA hybridization techniques, and Western blots.
- Protein engineering and protein design approaches; knowledge-based design, computational protein design, and directed evolution.
- Examples of engineered proteins with novel structural and functional properties
- Fermentation (basic principles of fermentation processes, scaling-up),
- Product recovery and separation (cell disruption, sedimentation, filtration/centrifugation, solvent extraction, adsorption, precipitation), product purification (adsorption, ultrafiltration, crystallization) and
- Final separation and advances in bioseparation strategies.
- Process design; creating engineered strains and cell culture/fermentation. Purification of the end product for energy, biotechnology or biomedical applications.